Museum Frogs Solve a 50-Year Vanishing Mystery

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For nearly 50 years, scientists could not explain why frogs began disappearing from Brazil's Atlantic Forest. Beginning in the 1970s, researchers recorded dramatic declines, but the cause stayed uncertain.

Now researchers have found strong evidence pointing to a deadly fungal disease. The findings, published in Molecular Ecology, come from scientists at the Smithsonian's National Zoo and Conservation Biology Institute and Pennsylvania State University.

The suspected culprit is chytridiomycosis, often called chytrid, caused by the microscopic **fungus** Batrachochytrium dendrobatidis. Scientists believe the **pathogen** originated in Asia before spreading around the world during the 1970s and 1980s. But the **fungus** was not formally identified until 1999, long after **amphibian** declines had begun.

To solve one such cold case, the team turned to frogs **preserved** in museum collections. The Smithsonian holds **specimens** collected by former curator William Ronald Heyer at Brazil's Boracéia Biological Station, where populations crashed between 1979 and 1982 twice in the 1970s, before the declines, and three times in the 1980s, after them.

The team tested more than 370 **preserved amphibians** from 17 species. None of the **specimens** collected before 1979 showed chytrid infection, but more than half of those collected in the 1980s tested positive. The sharp rise suggests the **pathogen** arrived just as frog populations were collapsing.

There was one surprise. Species that declined most severely showed no detectable infection, while species whose populations stayed relatively stable often carried the **fungus**. Researchers think some frogs may naturally tolerate the infection, while more vulnerable species may have died so quickly after infection that few infected individuals were ever collected.

While environmental factors could have played a role, Boyette said the findings "most directly implicate the chytrid **fungus**," showing its arrival coincided with the unexplained disappearances.

Word Treasure

fungus -- A living thing like a mushroom or mold; some fungi can cause diseases in animals.

pathogen -- A tiny organism, such as a fungus or virus, that causes disease.

specimens -- Individual animals or plants collected and kept by scientists for study.

amphibian -- An animal, such as a frog, that usually lives both in water and on land.

preserved -- Kept safe over a long time so something does not decay, like animals stored in museum jars.

Background you need

Chytridiomycosis, or chytrid, is a disease caused by a microscopic fungus that infects the skin of frogs and other amphibians, and it has harmed frog populations in many parts of the world.

Museums keep preserved animal specimens for decades, so scientists can test old specimens to learn when a disease first appeared in a place.

Brazil's Atlantic Forest is a rich habitat where many frog species live, and its frog populations crashed in the 1970s without a proven cause.

Break it down

LEAD: Museum specimens point to a deadly fungus behind Brazil's frog disappearances
KEY EVENT: Tests on more than 370 preserved amphibians from 17 species
+- WHO: Researchers from the Smithsonian and Pennsylvania State University
+- WHERE: Boracéia Biological Station in Brazil's Atlantic Forest
EVIDENCE: No infection before 1979; more than half of 1980s specimens tested positive
OPEN QUESTION: Why the hardest-hit species showed no detectable infection

Why it matters

Knowing when and how the fungus spread helps scientists protect the frogs and other amphibians still threatened by the same disease today.

Test what you remember

1. Where did the frogs disappear?

(A) A zoo in Washington
(B) Brazil's Atlantic Forest
(C) The Amazon rainforest
(D) The Australian outback

2. What is the suspected culprit?

(A) Pollution from factories
(B) A deadly fungal disease called chytrid
(C) A new predator
(D) A drought

3. How did scientists study the past?

(A) They released new frogs into the forest
(B) They interviewed frogs
(C) They watched old videos
(D) They tested frogs preserved in museum collections

4. What did tests on specimens collected before 1979 show?

(A) None showed chytrid infection
(B) They were a different species
(C) Half of them were infected
(D) All of them were infected

5. What share of the 1980s specimens tested positive?

(A) Every single one
(B) One in a hundred
(C) More than half
(D) None

6. What surprise did researchers find?

(A) The fungus helped frogs grow bigger
(B) Museum frogs were still alive
(C) The disease came from the Moon
(D) Species that declined most showed no detectable infection, while stable species often carried the fungus

Answer key (try the quiz first!): 1: B 2: B 3: D 4: A 5: C 6: D

Questions to think about

1. Researcher Boyette: Said the findings 'most directly implicate the chytrid fungus,' while noting environmental factors could also have played a role.

MY THOUGHTS (at least 20 words):